

Lab 5

(10+10+10+10+10 = 50 points total)

1. Using two registers, R0 and R1, and **conditional instruction(s) only**, write an assembly code to implement the function f defined as follows:

$$f(x) = \begin{cases} 0; & x < 0 \\ 1; & x \geq 0 \end{cases}$$

You can move some x to R0. The output of the function should be in R1.

2. Using two registers, R0 and R1, and **conditional instruction(s) only**, write an assembly code to implement the function f defined as follows (notice the difference between this problem and the previous one):

$$f(x) = \begin{cases} 0; & x \leq 0 \\ 1; & x > 0 \end{cases}$$

You can move some x to R0. The output of the function should be in R1.

3. Using two registers, R0 and R1, and **conditional instruction(s) only**, write an assembly code to implement the function f defined as follows (this function is known as the *rectified linear unit* (ReLU) in a neural network context):

$$f(x) = \begin{cases} 0; & x < 0 \\ x; & x \geq 0 \end{cases}$$

You can move some x to R0. The output of the function should be in R1.

4. Using two registers, R0 and R1, and **conditional instruction(s) only**, write an assembly code to implement the function f defined as follows:

$$f(x) = \begin{cases} -1; & x < 0 \\ x^2; & x \geq 0 \end{cases}$$

You can move some x to R0. The output of the function should be in R1.

5. Using two registers, R0 and R1, and **branch instruction(s) only**, write an assembly code to implement the function f defined as follows (the value of x can be interpreted as the day of the week (1 representing Monday) and the function outputs 1/0 can be interpreted as indicating whether it is a weekday or a weekend; note that the function definition is equivalent to a `switch` statement):

$$f(x) = \begin{cases} 1; & x = 1 \\ 1; & x = 2 \\ 1; & x = 3 \\ 1; & x = 4 \\ 1; & x = 5 \\ 0; & x = 6 \\ 0; & x = 7 \end{cases}$$

You can move some x to R0. The output of the function should be in R1.

Note: You can work on the lab assignment during the week and just demo it during the lab section next week (the week of May 25). Or you can choose to work on it during the lab hours next week. If you choose the latter option, please note that no grace period will be provided to finish your lab.